

Rocktim Jyoti Das

Brendan Irbe Center for Computer Science and Engineering,
University of Maryland, College Park, MD, USA

🏠 rocktimjyotidas.github.io
✉ rocktimjyotidas@gmail.com
🐙 GitHub & 🎓 Google Scholar

EDUCATION

University of Maryland, College Park

Aug, 2025 - May, 2030

PhD in Computer Science

Advisor: Prof. Dinesh Manocha

Indian Institute of Technology Delhi

Aug, 2018 - May, 2022

B.Tech in Electrical Engineering with Minor in Computer Science and Engineering

WORK EXPERIENCE

AMD Research and Advanced Development, Austin

June, 2026 - present

Research Associate - PhD

Mentor: Dr. Madhu Srinivasan

Physically Grounded Video Generation Model.

Mohamed bin Zayed University of Artificial Intelligence, Abu Dhabi

June, 2023 - Aug, 2025

Research Associate II

Advisors: Prof. Ivan Laptev & Prof. Preslav Nakov

Robotic Manipulation and Reasoning.

Data Analytics and Intelligence Research Group, IIT Delhi & IBM Research, India

June, 2022 - June, 2023

Research Assistant

Advisor: Prof. Mausam

Goal-oriented Dialogue Systems with Large Language Models

PUBLICATIONS

Conference Publications

1. **SLAT-Phys: Fast Material Property Field Prediction from Structured 3D Latents**
Rocktim Jyoti Das, Hyunjin Kim, Dinesh Manocha.
under review.
2. **EgoPhys: Physics-Informed Digital Twins for Deformable Objects from Egocentric Videos**
Hyunjin Kim, Ri-Zhao Qiu, Rocktim Jyoti Das, Guangqi Jiang, Dinesh Manocha, Xiaolong Wang
under review.
3. **BLAZER: Bootstrapping LLM-based Manipulation Agents with Zero-Shot Data Generation**
Rocktim Jyoti Das*, Harsh Singh*, Diana Turmakhan, Abdullah Sohail, Mingfei Han, Preslav Nakov, Fabio Pizzati, Ivan Laptev.
under review.
4. **MALMM: Multi-Agent Large Language Models for Zero-Shot Robotics Manipulation**
Harsh Singh*, Rocktim Jyoti Das*, Mingfei Han, Preslav Nakov, Ivan Laptev.
accepted in IROS (Oral) 2025.
5. **MALT: Improving Reasoning with Multi-Agent LLM Training**
Sumeet Ramesh Motwani, Chandler Smith, Rocktim Jyoti Das, Markian Rybchuk, Ronald Clark, Philip Torr, Fabio Pizzati, Ivan Laptev, Christian Schroeder de Witt
accepted in COLM 2025, ICLR SSI-FM 2025.
6. **EXAMS-V: A Multi-Discipline Multilingual Multimodal Exam Benchmark for Evaluating Vision Language Models**
Rocktim Jyoti Das, Simeon Emilov Hristov, Haonan Li, Dimitar Iliyanov Dimitrov, Ivan Koychev, Preslav Nakov.
Accepted in ACL Main, 2024.
7. **Beyond Size: How Gradients Shape Pruning Decisions in Large Language Models**
Rocktim Jyoti Das, Mingjie Sun, Liqun Ma, Zhiqiang Shen.
ArXiv 2024.
8. **Synergizing In-context Learning with Hints for End-to-end Task-oriented Dialog Systems**
Vishal Saley, Rocktim Jyoti Das, Dinesh Raghu, Mausam.
Accepted in EMNLP Main, 2024.

9. Exploring Distributional Shifts in Large Language Models for Code Analysis

Shushan Arakelyan, **Rocktim Jyoti Das**, Yi Mao, Xiang Ren.

Accepted in EMNLP Main, 2023.

Theses

1. Efficient Framework for Feature-Aware Graph Coarsening.

Rocktim Jyoti Das*, Aditi Khandelwal*, Sandeep Kumar, Jaydeva.

Best Undergraduate Thesis, Electrical Engineering, IIT Delhi, 2021 - 22.

AWARDS AND HONORS

- **Dean's Fellowship** awarded to selected first year students at University of Maryland, College Park 2025
- One of 12, of 20k applicants selected for **Google Predoctoral Research Program, 2023** 2023
- **Rajiv Bambawale Award** for Best Undergrad Thesis Award in EE Dept., IIT Delhi. 2022
- **Anundoram Borooah Award**, Government of Assam, India for academic excellence in high school. 2016

RESEARCH PROJECTS

Multi-sensory Object Physics Estimation

March 2026 – present

Advisors: Prof. Dinesh Manocha and Prof. Ruohan Gao

Developing a differentiable simulation framework to estimate material properties of rigid objects from audio and vision.

Piezoresistive Tactile Sensor

May 2026 – present

Advisor: Prof. Dinesh Manocha and Shengqiang Ren

Designing piezoresistive tactile sensors capable of measuring both shear and normal forces for robotic manipulation.

Learning Dynamics Models for Deformable Objects (Course Project)

Oct 2025 – Dec, 2025

Advisor: Prof. Ming C. Lin

Developed physical-ground digital twins by inferring material properties from real-world observation videos to enable accurate simulation and robotic manipulation of deformable objects.

SERVICE

- *Teaching Assistantship:*
 - Introduction to Computer Systems (UMD) - *Instr. Larry Herman* Spring 2026
 - Introduction to Data Science (UMD) - *Instr. Maksym Morawski* Fall 2025
 - Natural Language Processing (IITD) - *Prof. Mausam* Fall 2022
 - Advanced Machine Learning (IITD) - *Prof. Sandeep Kumar & Prof. Jaydeva* Spring 2022
 - Physical Electronics (IITD) - *Prof. Debanjan Bhowmik* Fall 2021
- *Journal Reviewer:* IEEE Robotics and Automation Letters (RA-L, 2025)
- *Conference Reviewer:* ICLR 2025, ICLR 2026, ICRA 2026, Neurips 2026, CoRL 2026, AAAI 2027
- *Organizer:* *Multimodal Reasoning Task* for ImageCLEF 2025 - Multimedia Retrieval Track at CLEF.
- *Outreach:* Mathematics tutor for high-school students at National Association of Blind, RK Puram, New Delhi.

SKILLS

- **Programming:** C++, MATLAB, Python and L^AT_EX
- **Framework:** PyTorch, PyTorch Geometric
- **Software:** CoppeliaSim, PyRep, RLBench, Issac-Sim
- **Robot:** Franka Emika Research 3

REFERENCES

- Dr. Dinesh Manocha, Professor, Department of Computer Science, UMD, Email: dmanocha@umd.edu.
- Dr. Ruohan Gao, Professor, Department of Computer Science, UMD, Email: rhgao@umd.edu.
- Dr. Ivan Laptev, Professor, Department of Computer Vision, MBZUAI, Email: ivan.laptev@mbzuai.ac.ae.
- Dr. Mausam, Professor, Computer Science and Engineering Department, IIT, Delhi, Email: mausam@cse.iitd.ac.in.
- Dr. Madhusudhanan Srinivasan, PMTS Software Development Engineer, AMD, Email: madhu.srinivasan2@amd.com.
- Dr. Christian Schroeder de Witt, RAEng Research Fellow, University of Oxford, Email: cs@robots.ox.ac.uk.